Simple Linear Regression

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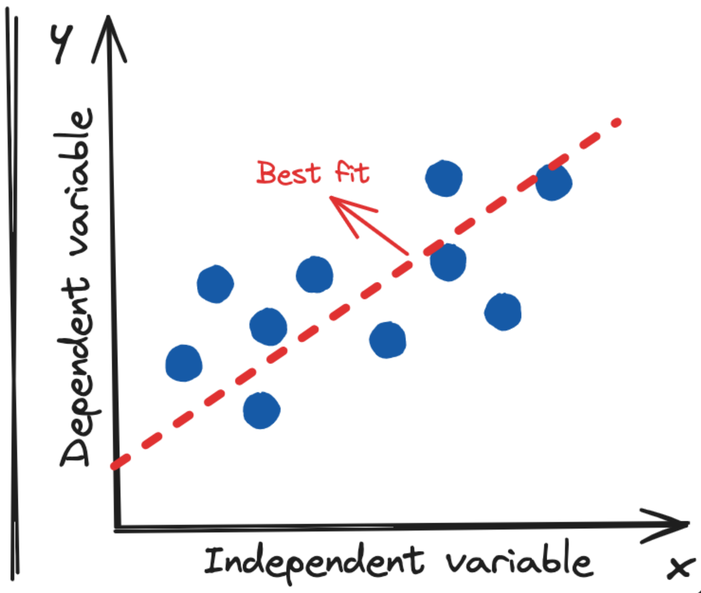
1. Visual Explanation

- Linear Regression is a simple statistical regression method, ideal for beginners.
- You can perform Linear Regression with multiple variables or just one.
  In this sheet, we use a single variable known as Simple Linear Regression.

↳ Goal: Find a line "Best fit" that represents the trend in the data.



2. Assumptions

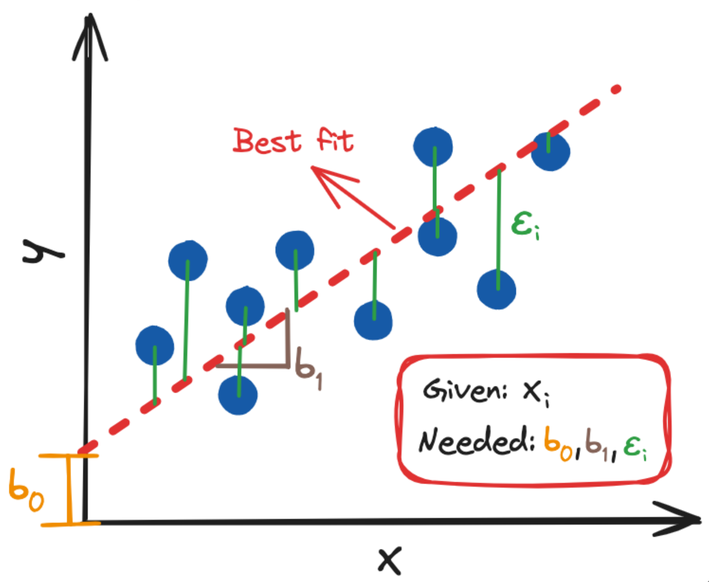
- Linearity: Linear Relationship between independent and dependent variables
- Independence: The residuals should be independent and uncorrelated.
- Homoscedasticity: The variance of the residuals must be constant.
- Normal distribution: The errors of the prediction follow a normal distribution with a mean close to zero.

3. Mathematical Explanation

- To calculate the best-fit line for the data, we use the following formula:

$$Y_i = b_0 + b_1 X_i + \varepsilon_i \qquad i = 1 \dots n$$

Y_i : Dependent Variable
b_0 : Intercept
b_1 : Slope
X_i : Independent Variable
ε_i : Random Errors (Residuals)

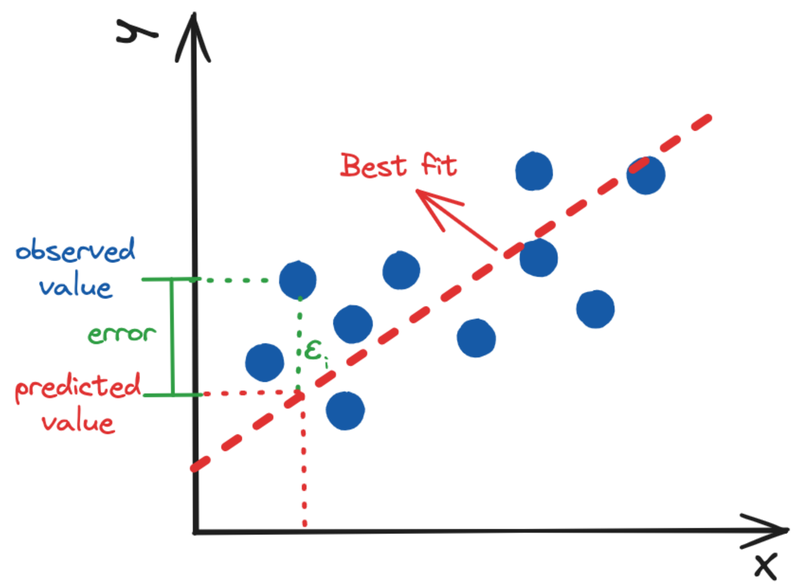


4. How can we calculate the Residuals?

- The best-fit line has the least error.
- The Residuals are the difference between the observed values of the dependent variable and the predicted values.

$$\varepsilon_i = y_i - y_i^{predicted}$$

↳ Goal: We have to minimize this error to get the best-fit line.



5. How can we calculate the Intercept and the Slope?

- We use the least squares method to calculate the intercept and the slope. This method is known as OLS regression (Ordinary Least Squares).
- We get the best fit line when the sum of the distance squares is minimal.

$$\sum_{i=1}^{n} (y_i - (b_0 + b_1 x_i))^2$$

- The least squares estimates are given by the following formulas:

$$b_0^{predicted} = \bar{y} - b_1^{predicted} \bar{x}$$

$$b_1^{predicted} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

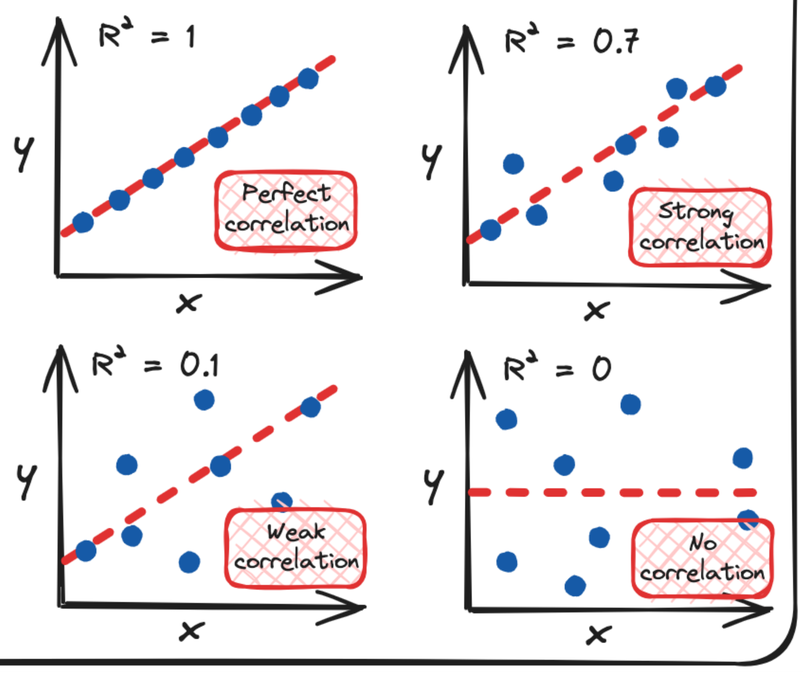
6. Evaluation

- Formula R-squared (R^2) or Coefficient of Determination:

$$1 - \frac{\sum_{i=1}^{n} (y_i - y_i^{predicted})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

Values between 0 and 1!

- The coefficient of determination describes the percentage of variance of the dependent variable (y) that can be explained by the independent variable (x).



7. Use of Linear Regression:

Pros

- Easy to interpret and understand
- Efficient method, relatively fast!
- Suitable for large datasets

Cons

- Statistical Assumptions
- Not suitable for complex data
- Sensitive to outliers

- Example: Increase in sales as a result of marketing activities

Discrete Distributions - Clearly explained

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Discrete distributions are statistical distributions used to model situations where the outcomes are distinct and countable, like the number of heads in a coin toss. They are defined by a probability mass function that assigns probabilities to these discrete outcomes.

Bernoulli Distribution

(Univariate Discrete)

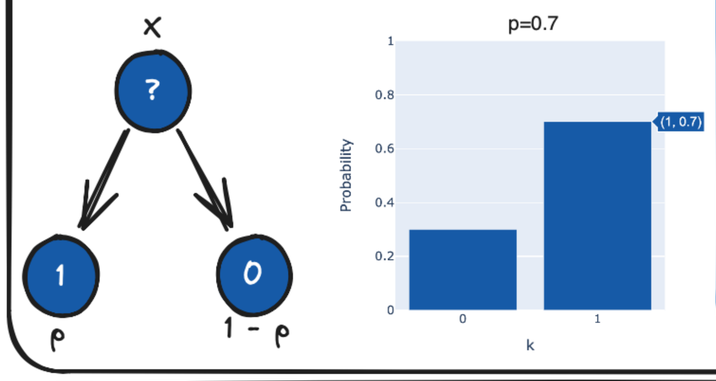
Notation:
$X \sim B(1,p)$, p $(0 \leq p \leq 1)$
Mean: p **Variance:** $p(1-p)$

Probability mass function (PMF):

$$p^X(k) = \begin{cases} p^k(1-p)^{1-k} & \text{for } k = 0,1 \\ 0 & \text{otherwise} \end{cases}$$

A Bernoulli distribution with parameter p exists if a random variable X has two possible outcomes (0 or 1). $X=1$ (success) occurs with a probability p, and $X=0$ (failure) occurs with a probability $1-p$.

Visual explanation:



Example:

- X represents the outcome of a coin toss, where $X=1$ (head) and $X=0$ (tail).
- p is the probability that the coin falls on head.

Binomial Distribution

(Univariate Discrete)

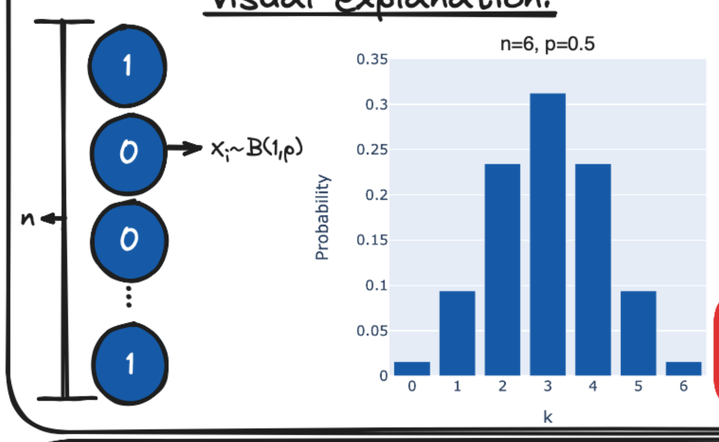
Notation:
$X \sim Bin(n,p)$, n,p $(0 \leq p \leq 1,\ n=1,2,\ldots)$
Mean: np **Variance:** $np(1-p)$

Probability mass function (PMF):

$$p^X(k) = \begin{cases} \binom{n}{k} p^k(1-p)^{n-k} & \text{for } k \in \{0 \ldots n\} \\ 0 & \text{otherwise} \end{cases}$$

A binomial distribution with parameters n and p exists if a random variable X is equal to the sum of n random variables that have a Bernoulli distribution with parameter p:
- $X = X_1 + X_2 + \ldots + X_n$
- X_i (Bernoulli distribution)

Visual explanation:



Example:

- X represents the total number of heads in n tosses of a coin.
- The probability of getting heads is p.

X -> total number of successes (number of 1s)

Poisson Distribution

(Univariate Discrete)

Distribution of rare events

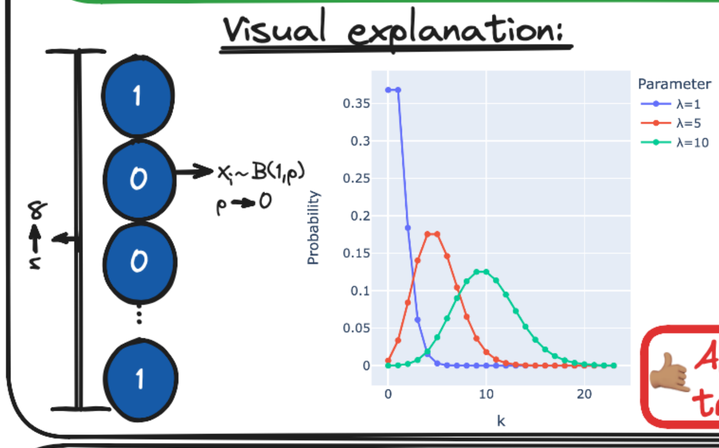
Notation:
$X \sim Poi(\lambda)$, λ $(\lambda > 0)$
Mean: λ **Variance:** λ

Probability mass function (PMF):

$$p^X(k) = \begin{cases} \frac{\lambda^k e^{-\lambda}}{k!} & \text{for } k = 0,1,2,\ldots \\ 0 & \text{otherwise} \end{cases}$$

The Poisson distribution is a limiting case of the Binomial distribution when the number of trials n tends to infinity and p tends to zero. The product $np=\lambda$ remains constant. λ indicates the average number of successful events.

Visual explanation:



Example:

A car insurance company needs the probability of 1, 2, or 3 cars having an accident in a year.

Approx. of the Binomial distribution: $n>=50$, $p<=0.05$

Geometric Distribution

(Univariate Discrete)

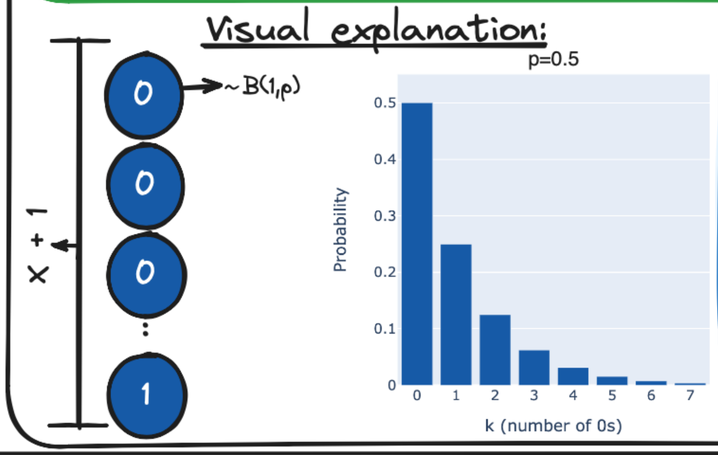
Notation:
$X \sim G(p)$, p $(0 \leq p \leq 1)$
Mean: $(1-p)/p$ **Variance:** $(1-p)p^2$

Probability mass function (PMF):

$$p^X(k) = \begin{cases} p(1-p)^k & \text{for } k = 0,1,2,\ldots \\ 0 & \text{otherwise} \end{cases}$$

We have a sequence of Bernoulli trials with parameter p. A random variable X with geometric distribution with parameter p represents the number of failed trials (number of 0s) that occur before the first success.

Visual explanation:



Example:

We can use the geometric distribution in determining the lifetime of equipment and components, i.e., waiting until the first failure.

Uniform Distribution

(Univariate Discrete)

Notation:
$X \sim U(a,b)$, a, b $(a, b \in \mathbb{Z})$
Mean: $(a+b)/2$ **Variance:** $\frac{(b-a+1)^2-1}{12}$

Probability mass function (PMF):

$$p^X(k) = \begin{cases} \frac{1}{b-a+1} & \text{for } k = a,a+1,\ldots b \\ 0 & \text{otherwise} \end{cases}$$

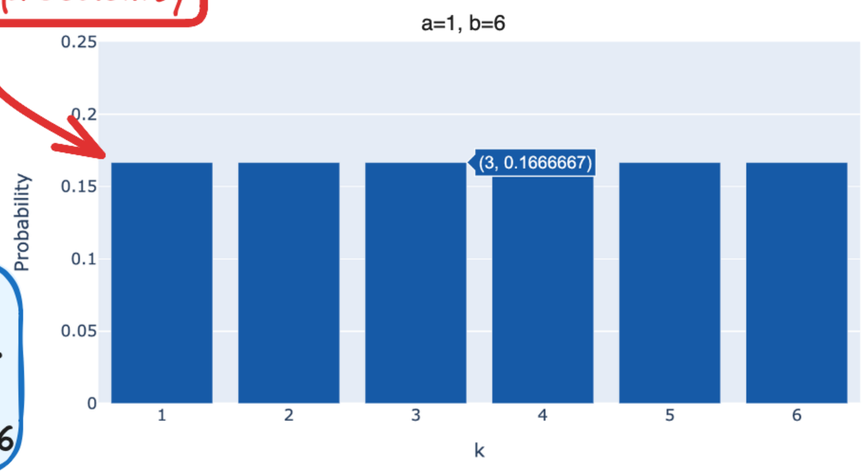
A random variable X with a discrete uniform distribution (parameters a and b) can assume any of the integers a to b with equal probability. All outcomes occur with the same probability in a uniform distribution.

Example:

A die is thrown once. The possible values of X are: $x_1=1, \ldots, x_6=6$

same probability

Visual explanation:



Cross Validation - Clearly explained

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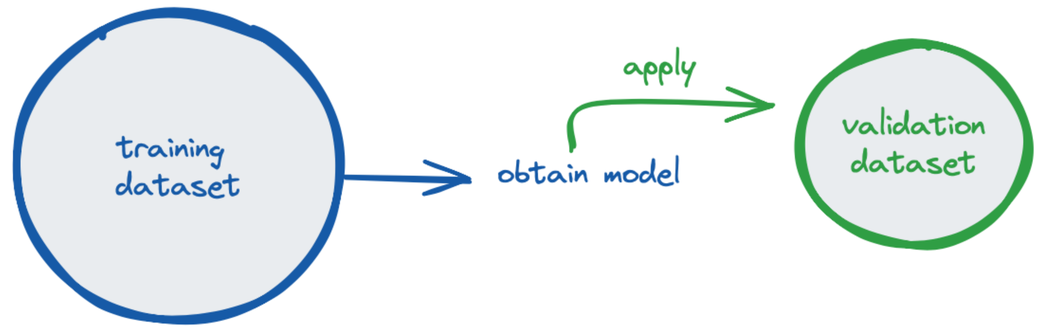
1. What is cross-validation?

Cross-validation (CV) is a way to test things in statistics by using resampling. It's an important tool today. Resampling means taking samples from a training dataset many times and fitting a model to each one.

2. Basic idea



- Large test datasets for evaluating statistical models are often unavailable
  → Cross-validation methods address this challenge
- CV's fundamental concept: Don't use entire dataset for fitting a model!
- Dataset is divided into training and validation datasets
- Validation dataset is typically smaller than the training set
- Fit a statistical model using the training dataset
- Apply the trained model to the validation dataset

3. Goodness of fit

Assess the goodness of fit by making predictions with the model and evaluating how well they align with the data.

There are three rates:

- Test error rate: Error in the prediction of test data
- Validation error rate: Estimated test error rate
- Training error rate: Error in the prediction of training data

Mean squared error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - y_i^{predicted})^2$$

n : Number of observations
y_i : class label for the i-th observation
$y_i^{predicted}$: predicted class label for the i-th observation

4. Validation Set Approach

Approach

- Validation set approach is the simplest CV type
- Divide dataset into training and validation datasets
- Fit model using training dataset
- Evaluate model's prediction on validation dataset

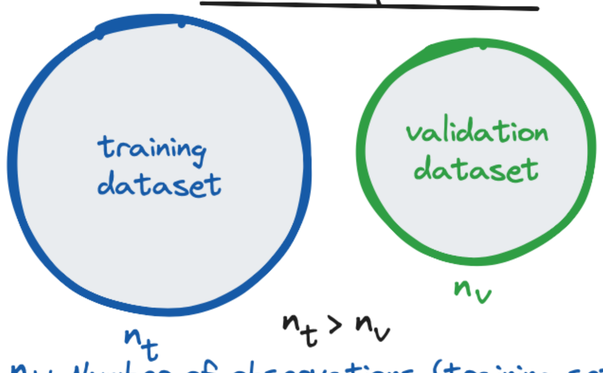
Advantages

- Simple strategy: Quick to execute

Disadvantages

- Strong dependence on distribution
- Fit the model only on the training dataset.

Visual Explanation



n_t: Number of observations (training set)
n_v: Number of observations (validation set)

Validation Error Rate

$$MSE = \frac{1}{n_v}\sum_{i=1}^{n_v}(y_i - y_i^{predicted})^2$$

n_v : Number of observations (validation set)
y_i : class label for the i-th observation
$y_i^{predicted}$: predicted class label for the i-th observation

5. Leave-One-Out Cross Validation (LOOCV)

Approach

- LOOCV divides the dataset into two parts
- A single observation is used as the validation dataset (validation data point)
- Remaining observations form the training dataset
- Each observation serves as the validation data point exactly once
- Model fitting and prediction of a validation data point are carried out n times

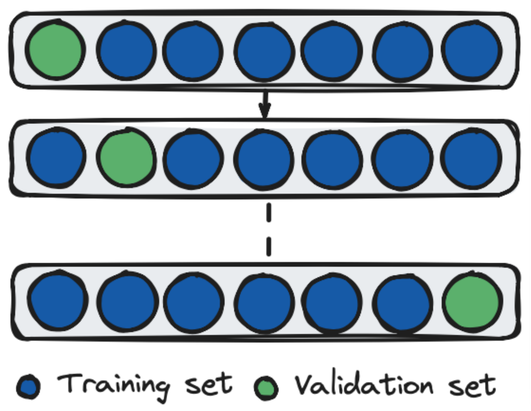
Advantages

- Using the entire dataset for training reduces test error rate overestimation compared to validation set approach.
- The split of the dataset is schematic.

Disadvantages

- High effort: We have to fit the model n times.

Visual Explanation



Training set Validation set

Validation Error Rate

$$MSE_i = (y_i - y_i^{predicted})$$
$$\Rightarrow CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} MSE_i$$

n : Number of observations (validation dataset)
y_i : class label for the i-th observation
$y_i^{predicted}$: predicted class label for the i-th observation

6. K-Fold Cross-Validation (k-fold CV)

Approach

- K-fold CV balances the validation set approach and LOOCV
- Randomly divides observations into k groups (folds) of roughly equal size
- Example: 3-fold cross-validation
  → First run: First group as validation dataset, others as training dataset
  → Second run: Second group as validation dataset
  → Third run: Third group as validation dataset

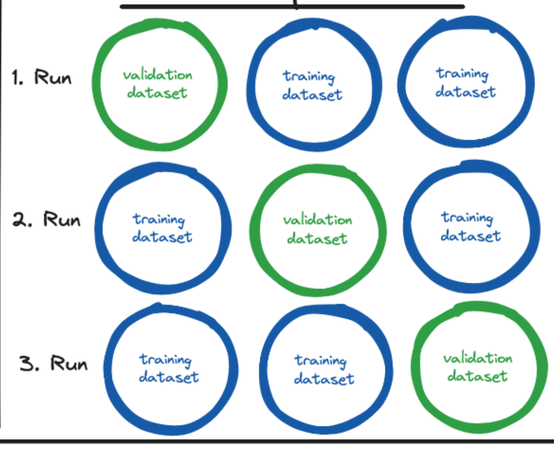
Advantages

- Less biased model than other methods
- It's one of the best methods if only limited input data is available.

Disadvantages

- We must fit the model k times. However, we can accept this disadvantage in order to estimate the test error rate as accurately as possible.

Visual Explanation



Validation Error Rate

$$CV_{(n)} = \frac{1}{k}\sum_{i=1}^{k} MSE_i$$

k: Number of groups/ folds

Useful resources:

- An Introduction to Statistical Learning

Ensemble Methods - Visually explained

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1. What are ensemble methods?

- An ensemble combines multiple models to create an even more powerful model.
- Assumption: The aggregation of multiple models mitigates the weaknesses of the individual models.

2. Motivation behind it

- Reducing the variance: Results are less dependent on the dataset.
- Reducing the bias: The combination of different models can learn different classes than individual models.
  - → Goal: Variance reduction without sacrificing bias!

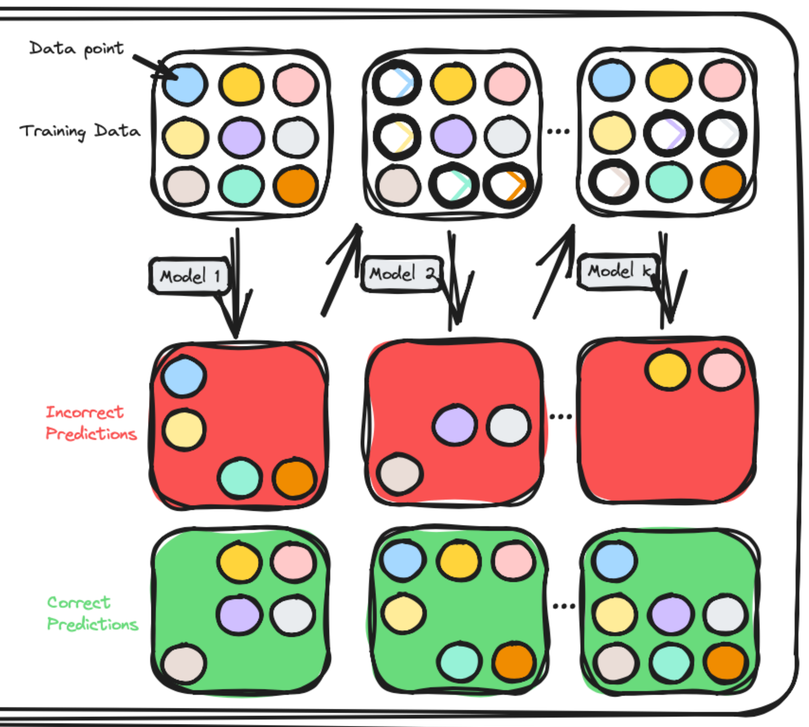
3. Strategies for ensembles

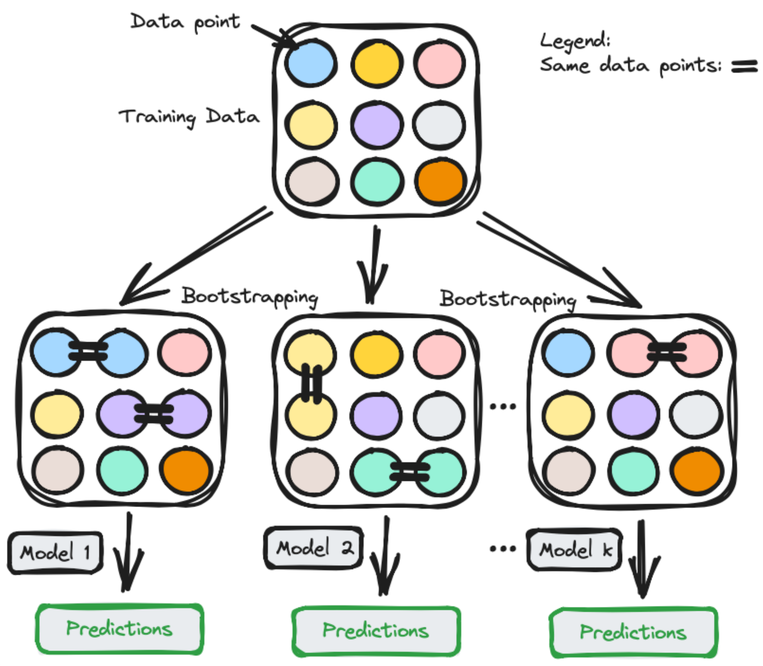
- Three strategies for ensembles:
  - → Boosting (Well-known)
  - → Bagging (Well-known)
  - → Stacking (not so well-known)

?

4. Boosting - How it works?

- Boosting involves going through an iterative training process.
- The subsequent model focuses more on the misclassified samples from the previous model.
- The final prediction is a weighted combination of all predictions.
- Examples: XGBoost, AdaBoost, and Gradient Boosting
- Many beginners struggles to understand how boosting works, so we created an easy-to-understand visual explanation.



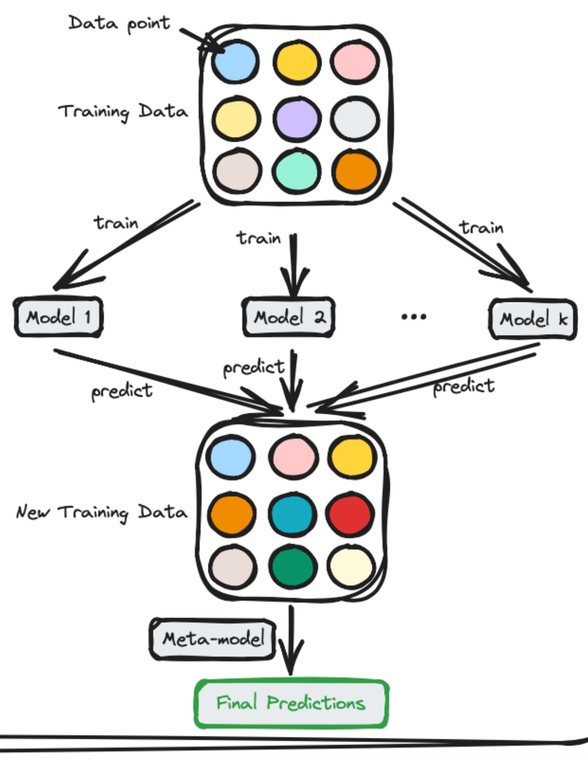


5. Bagging - How it works?

- Bagging (short for Bootstrapped Aggregation) creates different subsets of the data with replacement.
- Some data points may occur more than once in the individual subsets.
- We train one model with each subset. Then, we aggregate all predictions to get the final prediction.
- Example: Random Forest
- It can be difficult for beginners to understand the concept of bagging. So, take a look at the visual explanation.

6. Stacking - How it works?

- Stacking combines the predictions of multiple base models to achieve a final prediction.
- We train multiple base models on the same training dataset. Then, we feed the predictions into a meta-model to make a final prediction.
- Examples:
  - Base models:
    - → A wide range of models, for example Decision Tree, Neural Network, ...
  - Meta-model:
    - → Any algorithm, for example Neural Networks, ...



Prompt Engineering for Developers

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1. What is Prompt Engineering?

- It is a discipline for developing and optimizing prompts.
- Good prompts reveal LLMs' capabilities and limitations.

↳ Goal: Develop robust and effective prompting techniques.

2. What are the basic LLM types?

- Two main Types:
  - Base LLM: Trained on vast text data from the internet.
  - Instruction-tuned LLM: Fine-tuned on extensive data to follow instructions.

↳ Basis of an instruction-tuned LLM is a base LLM.

Base LLM

Prompt:

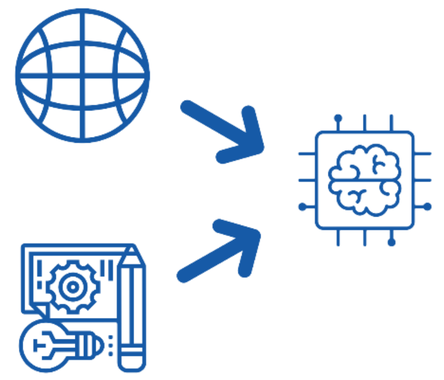
Who is the CEO of Tesla?

LLM Output:

Who is the CEO of Mercedes?

Who is the CEO of BMW?

Instruction-tuned LLM



Prompt:

Who is the CEO of Tesla?

LLM Output:

The CEO of Tesla is Elon Musk.

3. Clear and specific instructions

- Give the model clear and specific instructions
- Longer prompts often work better than short prompts.
  It provides more clarity and context for the model.

3.1 Use delimiters

- An effective technique to write effective prompts is to use delimiters.

Example:

```
text = f"""
Tesla Q1 2023 earnings report: \
Autopilot and Full Self-Driving (FSD) \
Our growing fleet of FSD Beta users has \
an exponential impact on total FSD Beta \
miles driven - with over 150 million miles \
to date and counting. This level of data \
collection is unprecedented in the industry. \
Mass collection of diverse datasets is essential \
for AI-based approach - the only approach \
we believe can work for scalable autonomy. \
In Q1, we enabled the latest FSD Beta \
software stack for highway driving.
"""
```

```
prompt = f"""
Summarize the text delimited \
by triple backticks into a \
single sentence.
```{text}```
"""
response = get_completion(prompt)
print(response)
```

Good

```
prompt = f"""
Summarize the text into a \
single sentence. {text}
"""
response = get_completion(prompt)
print(response)
```

BAD

3.2 Structured Output

- Ask for a structured output like HTML, XML, or JSON.

Example:

```
prompt = f"""
Generate a list of two company names along \
with their founding, CEO, and industry.
Provide them in JSON format with the following keys:
company_id, company_name, founding, ceo, industry.
"""
response = get_completion(prompt)
print(response)
```

LLM Output

```
[
  {
    "company_id": 1,
    "company_name": "ABC Corporation",
    "founding": "1995",
    "ceo": "John Smith",
    "industry": "Technology"
  },
  {
    "company_id": 2,
    "company_name": "XYZ Inc.",
    "founding": "2008",
    "ceo": "Jane Doe",
    "industry": "Retail"
  }
]
```

3.3 Few-shot-prompting

- Few-shot prompting means that we provide the model examples.
- It is very useful when you want to instruct the model to respond in a specific tone.

Example:

```
prompt = f"""
Your task is to answer in a consistent style.

Junior Analyst: Teach me about candle stick charts.

Senior Analyst: Candlestick charts are a popular tool \
in technical analysis for visualizing price movements \
in financial markets. Each candlestick represents a \
specific time period, showing the open and close \
prices as the rectangular "body" and the highest and \
lowest prices as "wicks" or "shadows." \
Different candlestick patterns can provide insights \
into market sentiment and potential future price \
movements, making them valuable for traders and investors.

Junior Analyst: Teach me about bollinger bands in three sentences.
"""
response = get_completion(prompt)
print(response)
```

LLM Output

Senior Analyst: Bollinger Bands are a technical analysis tool that consists of a moving average line and two standard deviation lines above and below it. They are used to measure volatility and identify potential overbought or oversold conditions in a market. When the price moves close to the upper band, it may indicate overbought conditions, while a move towards the lower band may suggest oversold conditions.

4. Divide and conquer

- Divide the task into meaningful steps.
- Give the model time to think.

Example:

```
prompt = f"""
Perform the following steps:
1 - Summarize the following text delimited by triple \
backticks with 2 sentence.
2 - Translate the summary from step 1 into German.
3 - What is the sentiment of the text, \
which is delimited with triple backticks?

Separate your answers with line breaks.

Text:
```{text}```
"""
response = get_completion(prompt)
print(response)
```

```
text = f"""
Tesla Q1 2023 earnings report: \
Autopilot and Full Self-Driving (FSD) \
Our growing fleet of FSD Beta users has \
an exponential impact on total FSD Beta \
miles driven - with over 150 million miles \
to date and counting. This level of data \
collection is unprecedented in the industry. \
Mass collection of diverse datasets is essential \
for AI-based approach - the only approach \
we believe can work for scalable autonomy. \
In Q1, we enabled the latest FSD Beta \
software stack for highway driving.
"""
```

5. Provide Reference Text

- Combat hallucinations by providing the LLM with reference texts.

↳ More accurate model answers!

- Example: Instruct the model to use specific texts and quotes from these texts.

Useful resource:

- OpenAI Prompt Engineering Guide

Return correlation in the portfolio example

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What is the best way to divide your money into different financial investments?

Investment returns

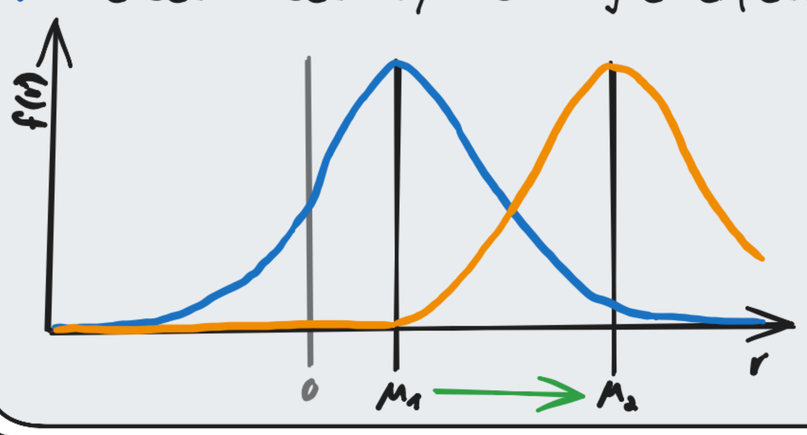
Investment returns are random variable R1, R2, ... (have an expected value, a variance, a distribution, etc.)

Expected return

- The security has a positive expected return.
- Calculate the loss probability as the area under the density.

Two securities: Expected returns

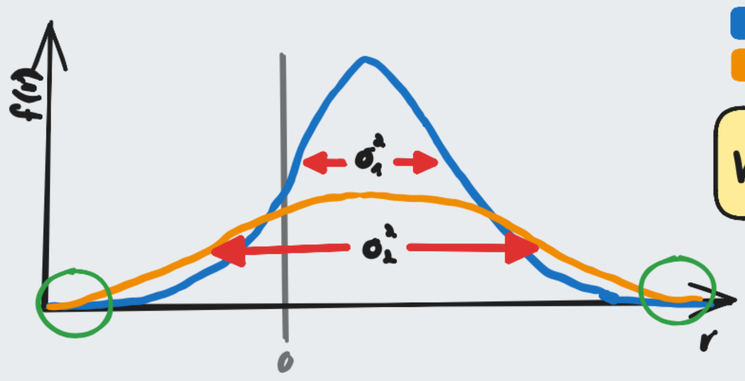
- The second security has a higher expected return.



$E[R_i] = \mu_i$: "Return"

Risk of the securities

- Both securities have the same expected value, but a different variance.

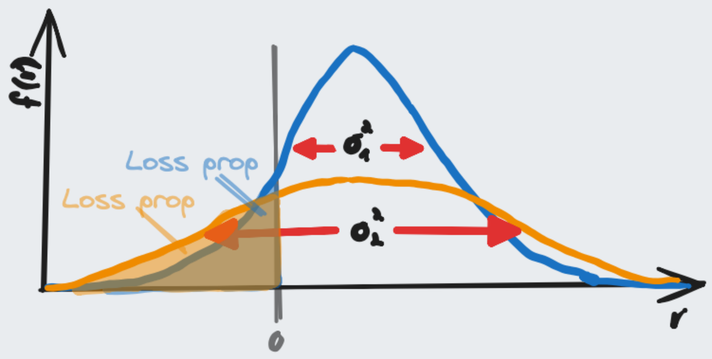


$VAR[R_i] = \sigma_i^2$: "Risk"

The return can be very high or very low. (marked in green)

Loss probability of the securities

- The loss probability is higher for the second security.



The second security is clearly riskier.

Portfolio allocation - Example with two securities

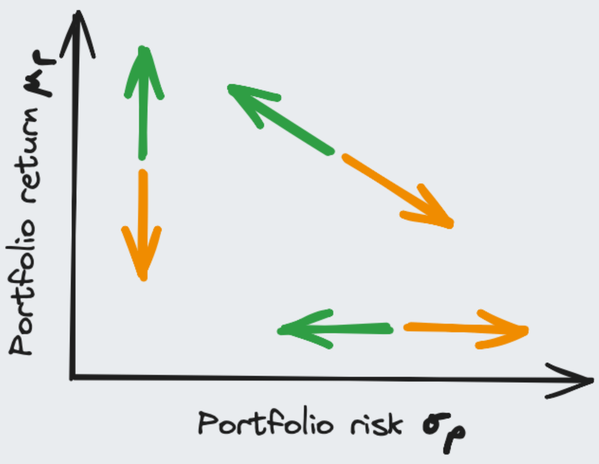
- The second security has a higher expected return (higher expected value), but also a higher risk (higher variance).
- The returns of the two securities are negative correlated. The development of the two returns is in opposite directions.
- Portfolio return: $R_p = gR_1 + (1-g)R_2$
  - g is the amount of the first security in the portfolio $0 \leq g \leq 1$
  - Portfolio return: $E[R_p] = \mu_p$
  - Portfolio risk: $VAR[R_p] = \sigma_p^2$

→ Portfolio return and Portfolio risk depend on the portfolio allocation g.

→ As a rule, the standard deviation rather than the variance is used directly for portfolio risk.

⚠ In reality, the correlation is typically not constant, as illustrated in our simple example.

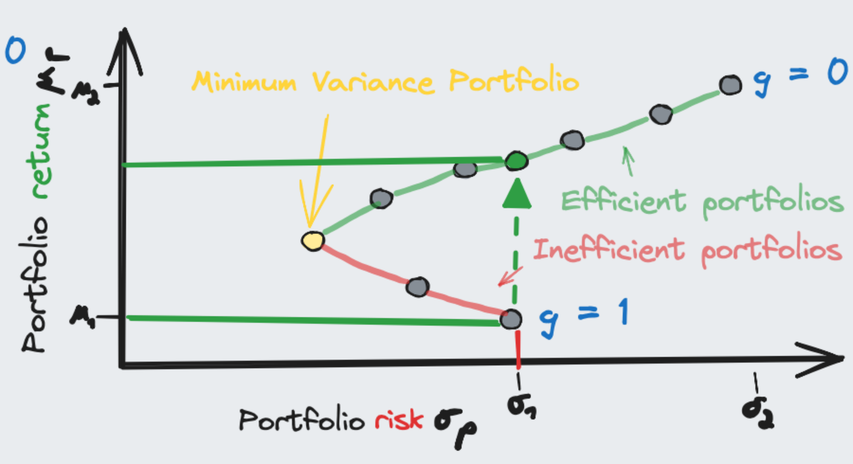
Mu-Sigma Diagram



- Visualize the optimal portfolio allocation with the Mu-Sigma diagram.
- The top left is always better than the bottom right.

$g = 1$: If we only hold the first security.

$g = 0$: If we only hold the second security. Higher return and higher risk than the first security.



- The possibility line illustrates the different allocations.
- Minimize the risk with a smart allocation.
- Minimum risk at the point marked in yellow.

By mixing the securities, we can reduce the portfolio risk compared to the individual risks (risk diversification).

YouTube Source: Korri der Korrelationskoala

Artificial Neural Networks

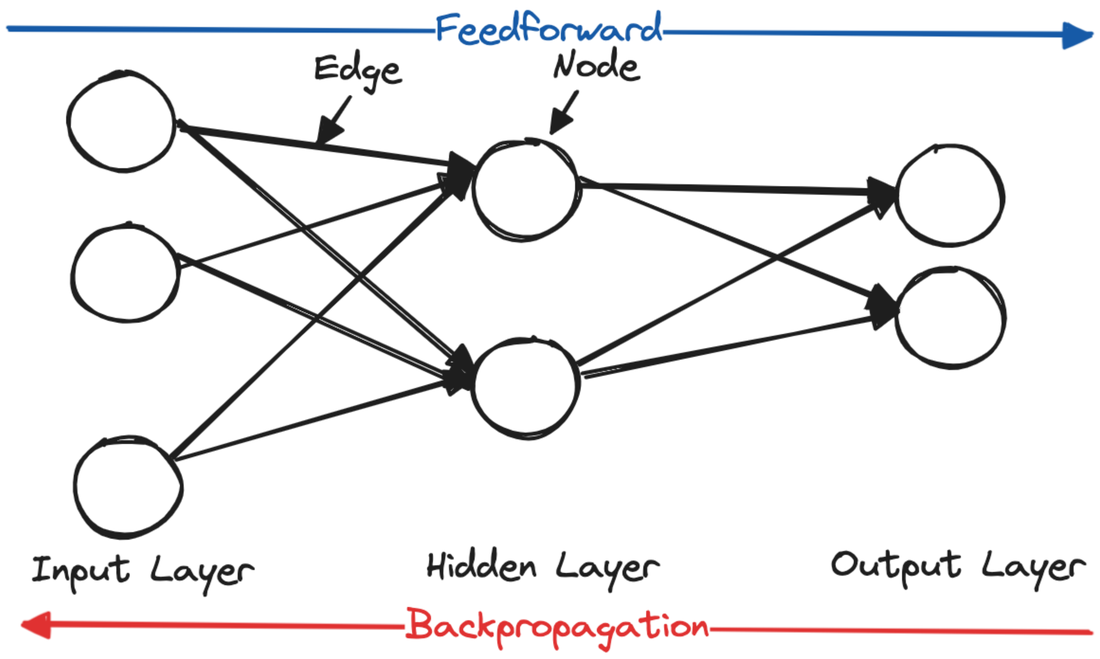
tinztwinshub.com

1. Neural Networks in General

- An Artificial neural network (ANN) consists of artificial neurons (nodes) and connections (edges).
- An ANN has one or more hidden layers, each layer consisting of several neurons.
- Neurons in each layer receive the output from neurons in previous layer as input.
- Each input to the neuron is weighted.
- The connections between nodes in this network are acyclic.

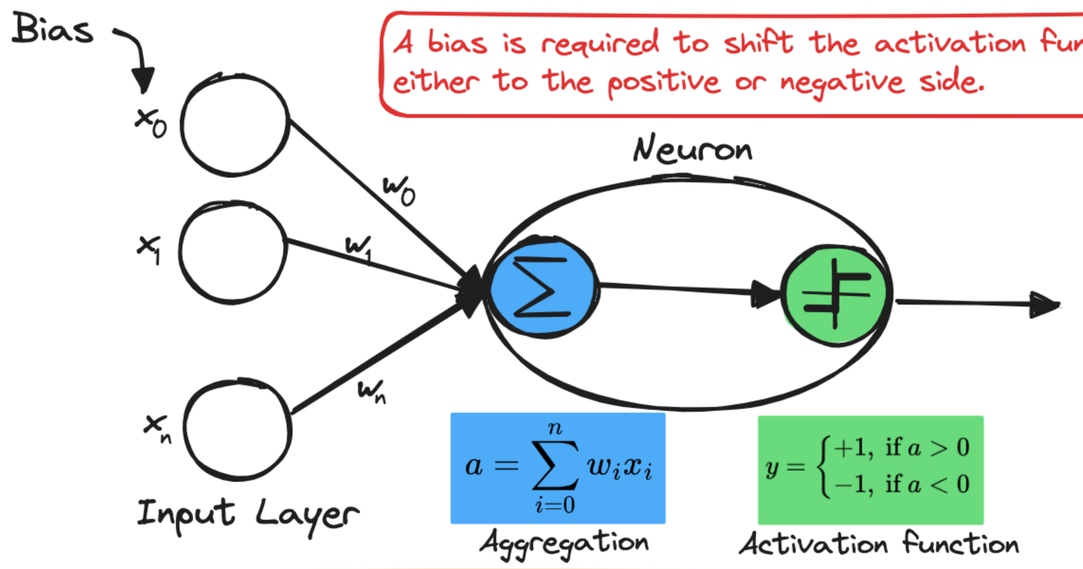
Feedforward is the flow of the input data through the ANN from the input layer to the output layer.

The return of the errors is called Backpropagation, where the errors are calculated per node.



2. Single-Layer Perceptron

- A perceptron consists of an artificial neuron.
- The neuron in the perceptron is called a Linear Threshold Unit (LTU).
- The neuron uses the step function as its output function and performs a linear separation of the input data.
- The input layer contains the input values and x_0 as bias.



Aggregation: The perceptron calculates the weighted sum of input values and weights.
Activation function: All values of a > 0 map to 1, and values a < 0 map to -1.

Limitations: The single-layer Perceptron can only solve linearly separable problems and struggles with complex patterns (e.g. XOR problem). The introduction of the multilayer perceptron and the backpropagation algorithm made it possible to solve nonlinear problems.

3. Feedforward in Detail

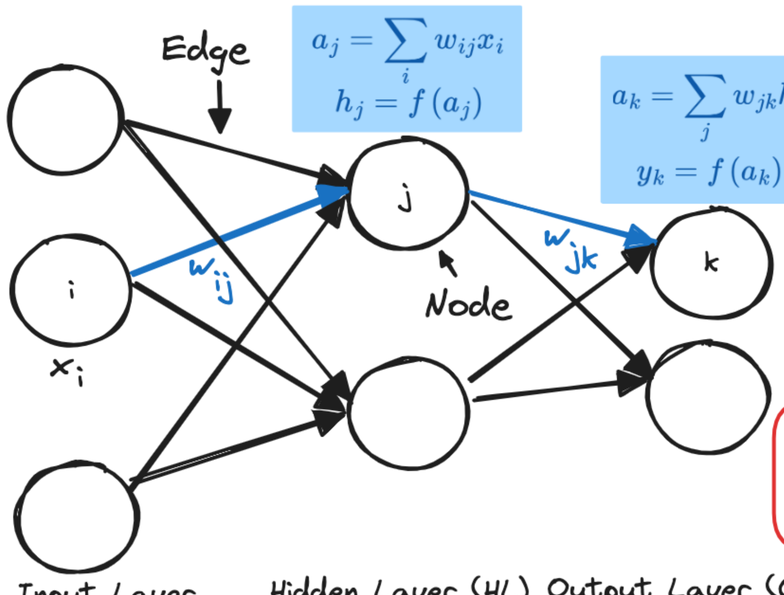
- In our example, we have exactly one hidden layer.
- The data x enters into the input layer. x is a vector with individual data points x_i.
- The individual data points are weighted with the weights w of the edges. There are different procedures for initializing the starting weights.

Calculations in the HL

The result of the aggregation is a_j and serves as the input of the activation function. Without an activation function, the output signal would just be a simple linear function.

Calculations in the OL

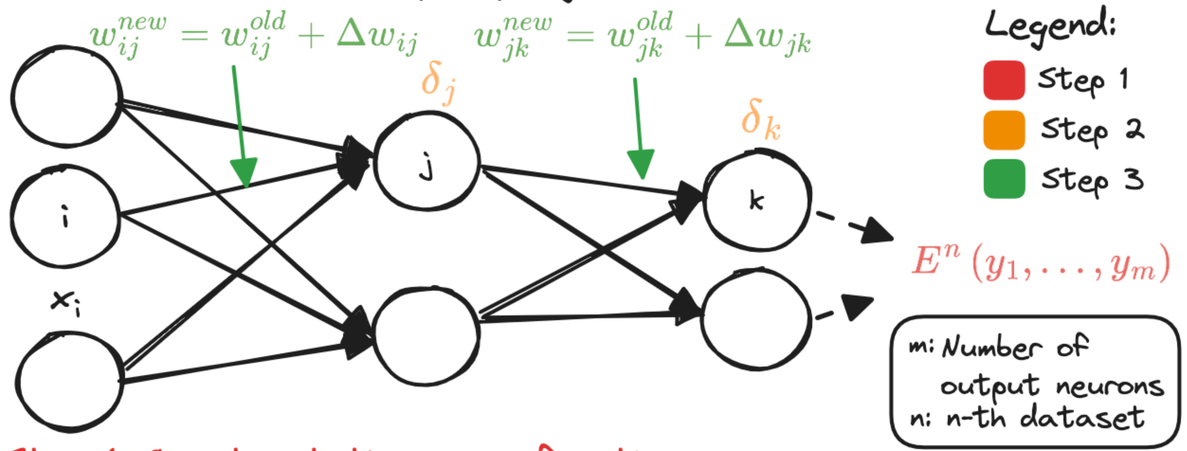
The results from the hidden layer serve as input for the output layer. A neuron in the output layer performs an aggregation. The result a_k is the input for the activation function.



a: Output of the aggregation
f: Activation function
h: Output hidden layer
y: Output vector of the network

Activation functions in the HL and OL can differ. Different functions are suitable for various use cases.

4. Backpropagation in Detail



Step 1: Construct the error function

The calculated values at the output of the network have an error compared to the true result values. An error function is used to calculate this error.

$$E^n(y_1, \ldots, y_m) = E^n(f(a_1), \ldots, f(a_m))$$

Step 2: Calculate the error terms for each node

To calculate the error terms for the output layer, we have to derive the error function according to the respective weights. For this, we use the chain rule.

$$\frac{\partial E^n}{\partial w_{jk}} = \underbrace{\frac{\partial E^n}{\partial a_k}}_{=:\delta_k} \cdot \frac{\partial a_k}{\partial w_{jk}} = \frac{\partial E^n}{\partial y_k} \cdot \frac{\partial y_k}{\partial a_k} \cdot \frac{\partial a_k}{\partial w_{jk}} = \underbrace{\frac{\partial E^n}{\partial y_k}}_{=:\delta_k} \cdot f'(a_k) \cdot h_j = \delta_k h_j$$

$$\delta_j = \frac{\partial E^n}{\partial a_j} = \sum_k \underbrace{\frac{\partial E^n}{\partial a_k}}_{=:\delta_k} \cdot \frac{\partial a_k}{\partial a_j} = \sum_k \delta_k \cdot \frac{\partial a_k}{\partial h_j} \cdot \frac{\partial h_j}{\partial a_j} = \sum_k \delta_k \cdot w_{jk} \cdot f'(a_j)$$

Step 3: Update of the weights on the edges

The new weights can calculate using the respective error terms and a learning rate (Eta sign). The learning rate specifies how strongly the weights are adjusted.

$$\Delta w_{jk} := -\eta \delta_k h_j$$
$$\Delta w_{ij} := -\eta \delta_j x_i$$

Time Shifting in Pandas - Clearly explained

tinztwinshub.com

For certain Machine Learning models, it's necessary to shift the time series data by a specific number of steps. The Pandas Library excels at this task. Whether you need to shift your time series data forward or backward, the shift() function can handle it.

1. Pandas shift() function:

```
DataFrame.shift(periods=1, freq=None, axis=0, fill_value=_NoDefault.no_default)
```

Let's quickly go over the parameters:

- periods: Number of periods to shift. It can be either positive or negative.
- freq: Frequency-based shifting. The DataFrame's index must be a date or datetime; otherwise, an error will occur.
- axis: Shift direction (0 or 'index', 1 or 'columns')
- fill_value: You can select the fill value, such as 0.

2. Shifting time series with periods

Want to shift your data by a specific period?

- In the example, we set periods=1, which means we shift the time series forward by one day.
- The row values are moved forward by one day (marked in red), so the first row would contain NaN.

But how can we fill in the missing values with a specific value?

- We can use the fill_value parameter to define which value replaces the missing values. For example: 0 (marked in blue)

Time series data of the Tesla stock

date	open	high	low	close	volume
2023-01-03	118.47	118.80	104.64	108.10	231402800
2023-01-04	109.11	114.59	107.52	113.64	180389000
2023-01-05	110.51	111.75	107.16	110.34	157986300
2023-01-06	103.00	114.39	101.81	113.06	220911100
2023-01-09	118.96	123.52	117.11	119.77	190284000

↓ `df_daily.shift(periods=1, fill_value=0)`

date	open	high	low	close	volume
2023-01-03	0.00	0.00	0.00	0.00	0
2023-01-04	118.47	118.80	104.64	108.10	231402800
2023-01-05	109.11	114.59	107.52	113.64	180389000
2023-01-06	110.51	111.75	107.16	110.34	157986300
2023-01-09	103.00	114.39	101.81	113.06	220911100

3. Shifting time series with freq

The freq parameter lets you shift the time series based on frequency. In this scenario, we shift the index.

As a result, the entire time series data has been moved forward by three days. Additionally, there are no NaN values because we only shifted the index.

Time series data of the Tesla stock

date	open	high	low	close	volume
2023-01-03	118.47	118.80	104.64	108.10	231402800
2023-01-04	109.11	114.59	107.52	113.64	180389000
2023-01-05	110.51	111.75	107.16	110.34	157986300
2023-01-06	103.00	114.39	101.81	113.06	220911100
2023-01-09	118.96	123.52	117.11	119.77	190284000

↓ `df_daily.shift(freq='3D')`
or
`df_daily.shift(periods=3, freq='D')`

date	open	high	low	close	volume
2023-01-06	118.47	118.80	104.64	108.10	231402800
2023-01-07	109.11	114.59	107.52	113.64	180389000
2023-01-08	110.51	111.75	107.16	110.34	157986300
2023-01-09	103.00	114.39	101.81	113.06	220911100
2023-01-12	118.96	123.52	117.11	119.77	190284000

4. Difference between the closing prices of two consecutive days

We can calculate the change in closing prices from one day to the next with shift().

Python Code: `df_daily['difference'] = df_daily['close'] - df_daily.shift(1)['close']`

date	open	high	low	close	volume	difference
2023-01-03	118.47	118.80	104.64	108.10	231402800	NaN
2023-01-04	109.11	114.59	107.52	113.64	180389000	5.54
2023-01-05	110.51	111.75	107.16	110.34	157986300	-3.30
2023-01-06	103.00	114.39	101.81	113.06	220911100	2.72
2023-01-09	118.96	123.52	117.11	119.77	190284000	6.71

In the output dataframe, we can see the price change in the difference column, highlighted in orange. The first value in this column is NaN because there is no previous row to compare it with.

What is CRISP-DM?

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CRISP-DM stands for CRoss Industry Standard Process for Data Mining. The model breaks a data science or data mining project into distinct, clearly defined steps. Each step focuses on specific tasks. CRISP-DM includes six process phases that can be repeated multiple times.

Business Understanding

- At the start of a Data Science project, the focus is on understanding the use case.
- All stakeholders define the common goal and acceptance criteria for evaluation.
- Most people underestimate the importance of business understanding. Please don't do that!
- All stakeholders must work towards the common goal.
- The result of this step is a concrete project plan.

Data Understanding

- In this phase, you examine the quality of the available data.
- You can use methods from exploratory data analysis to understand the data.
- The goal of data understanding is to check the feasibility of the project.

Data Preparation

Data preparation involves creating a dataset for the next phases. You prepare and clean the dataset.

This phase includes three tasks:

Merging the data | Data cleaning | Feature engineering

Ensure that you merge data from different sources meaningfully. This step is the foundation for data cleaning and feature engineering.

Modeling

Core of CRISP-DM

- You use Machine Learning (ML) methods to develop an ML model based on the prepared historical data.
- Usually, you divide the data into test and training data.
- You train the ML model with the training data and evaluate it with the test data.

The goal is to achieve the highest possible prediction rate.

Evaluation

- The data science team presents the results from the previous two steps to the stakeholders.
- Stakeholders analyze the overall process from data understanding to modeling.
- They decide whether the developed solution is ready to operationalize.

Deployment

- You integrate the developed ML solution into the existing infrastructure.
- It is important to continuously monitor the ML models.

Limitations of CRISP-DM

There are some gaps in the CRISP-DM model. The following points are not included in the CRISP-DM standard:

Data acquisition

- The CRISP-DM assumes that the data is already available.
- In practice, however, the data is often not available or only excerpts are available.

No project management

- The CRISP-DM model does not include project management.
- You need to use other methods for project management.
- Scrum is an established method in software development.
- You can combine the Scrum framework with CRISP-DM.

SQL Joins Visualization

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Table A

a_id	ticker
1	AAPL
2	GOOG
3	TSLA
4	SQ

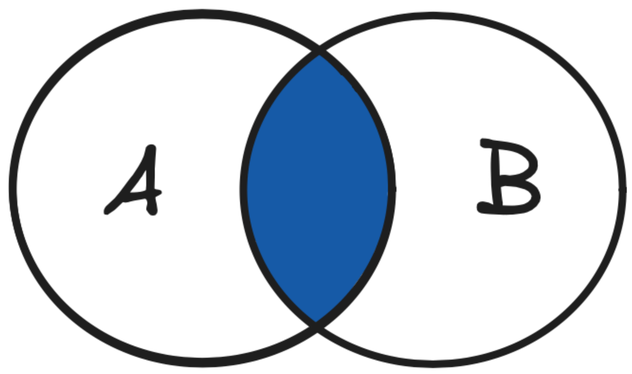
Table B

b_id	ticker
A	AAPL
B	PLTR
C	MSFT
D	TSLA

Order of execution

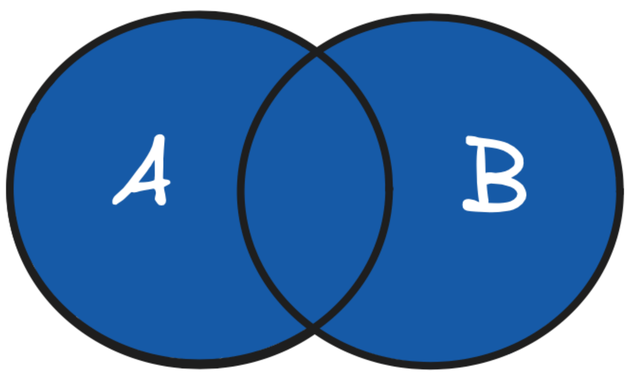
1. FROM/JOIN
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. ORDER BY
7. LIMIT

Inner Join



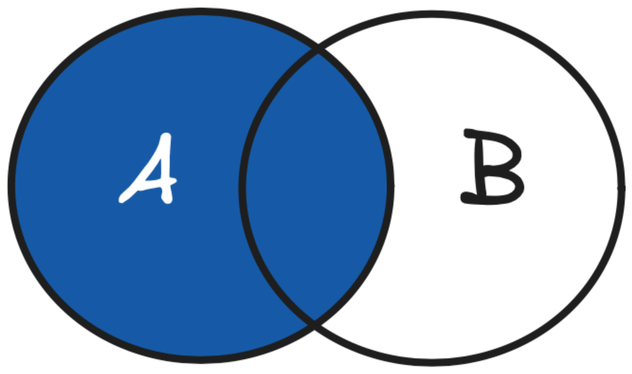

```
SELECT *
FROM A
INNER JOIN B
  ON A.ticker = B.ticker
```

Full Outer Join



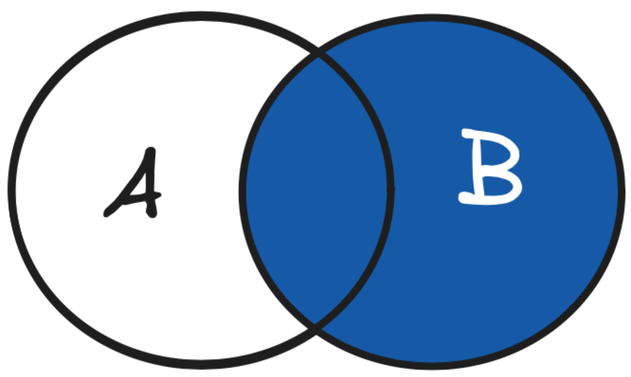

```
SELECT *
FROM A
FULL OUTER JOIN B
  ON A.ticker = B.ticker
```

Left Outer Join



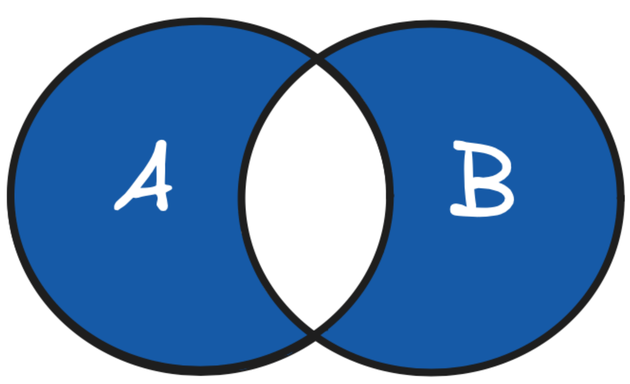

```
SELECT *
FROM A
LEFT OUTER JOIN B
  ON A.ticker = B.ticker
```

Right Outer Join



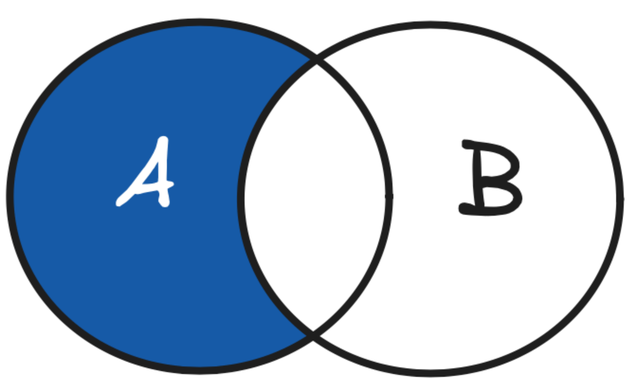

```
SELECT *
FROM A
RIGHT OUTER JOIN B
  ON A.ticker = B.ticker
```

Outer Excluding Join



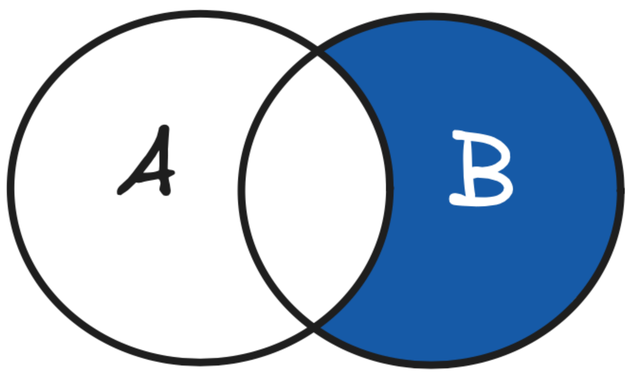

```
SELECT *
FROM A
FULL OUTER JOIN B
  ON A.ticker = B.ticker
WHERE A.ticker IS NULL
  OR B.ticker IS NULL
```

Left Excluding Join




```
SELECT *
FROM A
LEFT JOIN B
  ON A.ticker = B.ticker
WHERE B.ticker IS NULL
```

Right Excluding Join




```
SELECT *
FROM A
RIGHT JOIN B
  ON A.name = B.name
WHERE A.name IS NULL
```

Docker - Clearly explained

tinztwinshub.com

Docker is a tool for simplifying application development and deployment. It makes it easy to package, distribute, and run applications using images and containers.

Dockerfile

What it is:

A file that includes all the steps to create a Docker image.

Explanation:

It's like a recipe for creating your Docker image. It specifies the base image, adds your application code, sets configurations, and defines how the image should be constructed.

Docker Image

What it is:

Think of a Docker image as a ready-to-go template of your application.

Explanation:

It contains everything your application needs to run: source code, libraries, dependencies, operating system.
You can create an image with a Dockerfile.

Docker Registry

What it is:

A centralized repository for Docker images.

Explanation:

The Docker Registry (like Docker Hub or GitLab Container Registry) is where you store and share your Docker images. It's like an online library for images that you and others can use.

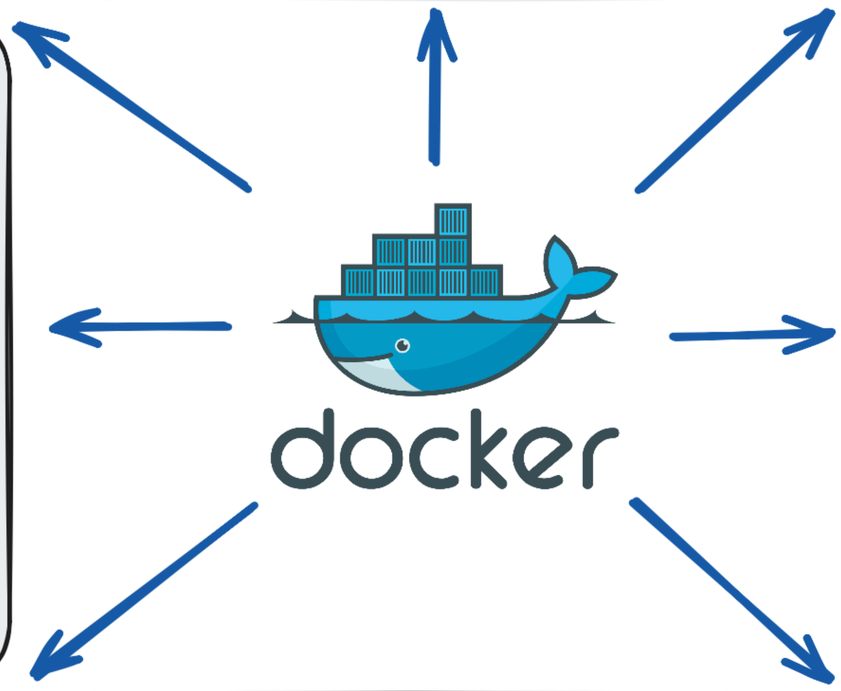
Docker Container

What it is:

A running instance of a Docker image.

Explanation:

If an image is a template of your application, a container is the running application. It is isolated from the rest of your system, ensuring that it works on any computer system.



Docker Volume

What it is:

A way to persist data outside of a container.

Explanation:

Containers are fleeting. Volumes provide a way to store and share data between your host machine and the container, ensuring that your data is not lost when you stop or delete the container.

Docker Network

What it is:

A communication bridge between Docker containers.

Explanation:

Docker Network allows containers to talk to each other. It's like a private network for your containers, ensuring seamless communication while keeping them isolated from external influences.

Docker Daemon

What it is:

The background service that manages Docker objects on your system.

Explanation:

The Docker Daemon is like a behind-the-scenes worker. You interact with it through the Docker command-line interface (CLI). Docker objects include images, containers, volumes, and networks.

Docker Engine

What it is:

The combination of the Docker Daemon and the Docker CLI.

Explanation:

The Docker Engine is the overall software that allows you to work with containers. You use the Docker CLI to give commands, and the Docker Daemon manages the containers.

Check your Docker Version:
`docker --version`

Essential Docker Commands

Docker Image

```
docker pull <image-name>
docker run -d <image-name>
docker rmi <image-name>
docker image list
docker image prune
```

Docker Container

```
docker ps -a
docker stop <container-id>
docker restart <container-id>
docker rm <container-id>
docker exec -it <container-id> bash
docker container prune
```

Docker Volume & Network

```
docker volume create <volume-name>
docker volume rm <volume-name>
docker volume ls
docker network create <net-name>
docker network rm <net-name>
docker network ls
```

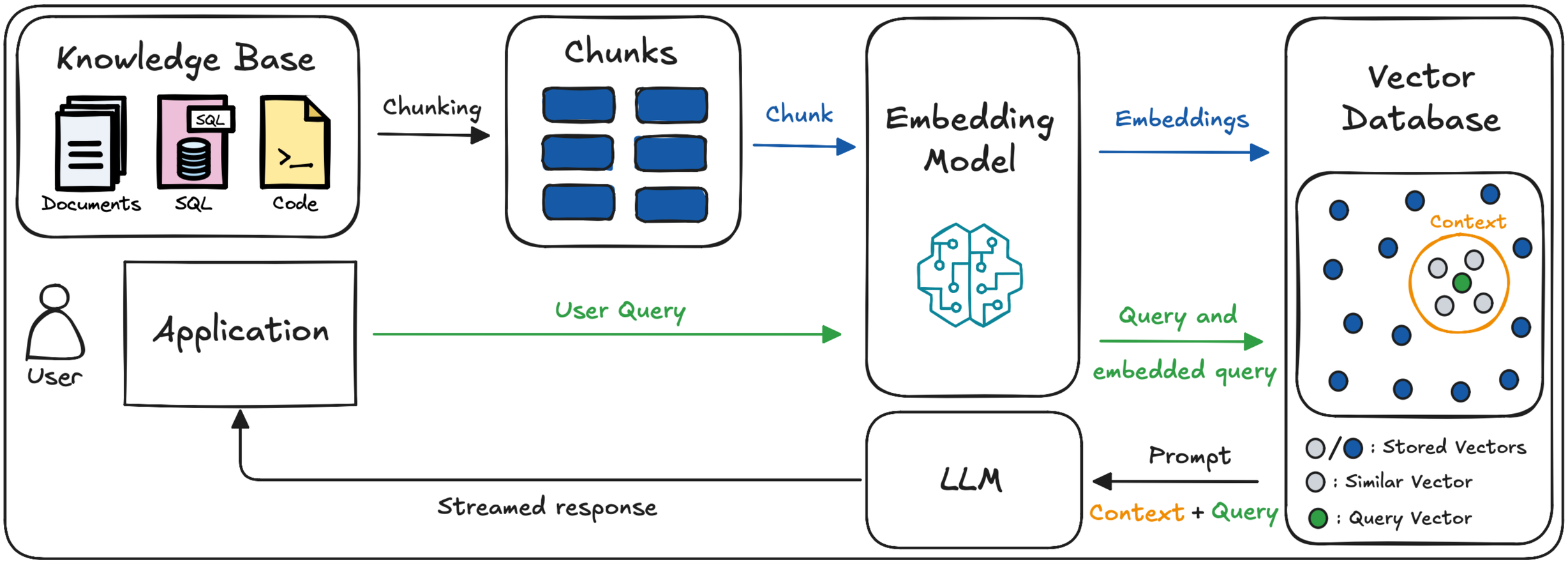
What is Retrieval-Augmented Generation?

tinztwinshub.com

Retrieval-Augmented Generation (RAG) is an advanced AI technique that improves LLM responses. The RAG system's retriever can access relevant information from external sources, such as documents, web pages, databases, or other sources. That means the LLM is not limited to its internal knowledge.

RAG Workflow

1. The user enters a query.
2. Retriever fetches information relevant to the query.
3. Relevant information is augmented to the prompt as context.
4. LLM generates a response to the prompt
5. Response to the user



Use Cases

Document Question Answering Systems

- Generating precise, contextually relevant answers

Conversational Agents

- RAG improves interaction with the user.
- Generating context-specific responses

Personalized Recommendation

- RAG improves this task by creating tailored and contextually appropriate suggestions for users.

Content generation

- RAG systems produce contextually accurate and comprehensive content.

Pros and Cons of RAG

Pro

- Context Awareness: Improves the accuracy
- Source Citation: Access to sources enhances the transparency of the LLM response.
- Reduction of Hallucinations
- Good for domain-specific data

Contra

- Latency: Extended response time can be problematic in real-time applications.
- An external knowledge base is required.